

# CareAI methodology comparison and strongest path

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## CareAI methodology comparison and strongest path

The literature points to a clear judgment. The strongest path for CareAI is not a plain HAN++ paper that only claims better scores than older baselines. The strongest path is a harder and more modern paper in which HAN++ remains the relational core, but is tested against strong sequential, transformer, graph transformer, hypergraph, and knowledge-augmented baselines under a much stricter validation package [Che24b, Jia23b, Xu23, Li19, Ras20, Cho16b, Liu20].

A plain HAN++ story is still publishable if the gains are strong. It is no longer the strongest path. Recent EHR work has moved toward temporal heterogeneous graphs, pretraining, relation-aware transformers, and personalized knowledge graphs [Che24b, Hua24, Jia23b]. At the same time, methodology papers make clear that many model papers overclaim because they rely on retrospective benchmarks, weak calibration analysis, and limited external validation [Alb17, Ril21, Col15c, Wol19].

## Judgment first

The best judged path for CareAI is this:

- Keep HAN++ as the interpretable heterogeneous relational backbone
- Add explicit temporal modeling across visits
- Stress-test graph construction and meta-path choices
- Compare against strong modern baselines, not just older RNN or shallow models
- Make the paper win on validation quality as much as on model design

That path is stronger than three alternatives.

| Path                                                                   | Judgment                    | Why                                                                                                      |
|------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------|
| Plain HAN++ against older baselines                                    | Weakest of the viable paths | Too easy to challenge with newer graph, transformer, and pretraining papers [Che24b, Hua24, Jia23b]      |
| HAN++ plus strong validation only                                      | Better                      | Can produce a solid clinical methods paper, but may still lose the architecture argument to newer models |
| HAN++ plus temporal layer plus hard baselines plus clinical validation | Strongest near-term path    | Best balance of novelty, feasibility, and defensible claims [Che24b, Liu20, Li19, Ras20]                 |
| Full shift to a new foundation model or graph transformer              | High upside but riskier     | Potentially stronger technically, but farther from the model the team already has [Hua24, Hu20]          |

## Comparison of methodologies across representative studies

| Study             | Study design and data view                                                      | Core technique                               | Analytical framework                                              | What it means for CareAI                                                                                    |
|-------------------|---------------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| [Cho15] Doctor AI | Retrospective longitudinal EHR as visit sequence                                | RNN for next-visit multi-label prediction    | Sequence modeling of coded visits                                 | Strong classic temporal baseline. Good for proving HAN++ adds more than sequence memory                     |
| [Cho16b] RETAIN   | Retrospective longitudinal EHR with clinician-facing interpretation goal        | Reverse-time two-level attention             | Visit-level and code-level attention over sequences               | Important interpretability baseline. If HAN++ cannot beat or complement this, the explanation claim is weak |
| [Cho16] GRAM      | Retrospective EHR with ontology support                                         | Attention over medical ontology ancestors    | Knowledge-guided sequence representation                          | Strong when rare codes and medical hierarchy matter. Useful non-graph knowledge baseline                    |
| [Ma17] Dipole     | Retrospective longitudinal EHR                                                  | Bidirectional RNN with attention             | Temporal dependency plus visit attention                          | Good strong pre-transformer sequential baseline                                                             |
| [Li19] BEHRT      | Large-scale retrospective EHR                                                   | Transformer over structured records          | Self-attention sequence model with demographic and visit encoding | Hard baseline for multi-label disease prediction at scale                                                   |
| [Ras20] Med-BERT  | Large-scale pretrain then fine-tune on structured EHR                           | BERT style pretraining                       | Foundation model transfer learning                                | Critical baseline if CareAI data are not huge. Pretraining can erase gains from custom architecture         |
| [Hos18] HeteroMed | Retrospective single-admission heterogeneous medical graph                      | Metapath-based heterogeneous graph embedding | Static heterogeneous graph with supervised diagnosis objective    | Early justification for graph heterogeneity, but weak on time and external validity                         |
| [Hos19] HTAD      | Retrospective heterogeneous graph with mixed categorical and time-series inputs | Target-aware hierarchical attention          | Diagnosis-specific node and type weighting                        | Closest ancestor to a HAN++ style clinical story. Good support for targeted heterogeneous attention         |

| Study                         | Study design and data view                                                     | Core technique                                         | Analytical framework                                                           | What it means for CareAI                                                                            |
|-------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| [Qu22] and [Qu23]             | Retrospective heterogeneous EHR graph                                          | HAN and metapath-aggregated HAN for disease prediction | Node-level and semantic-level attention over metapaths                         | Direct support for HAN style modeling in EHR. Useful but not sufficient as strongest comparator set |
| [Liu20] HSGNN                 | Retrospective heterogeneous EHR graph                                          | Hub-aware heterogeneous similarity graph fusion        | Meta-path decomposition with PathSim normalization                             | Important challenge to HAN++. Shows hub nodes can break naive heterogeneous attention               |
| [Cho19] GCT                   | Retrospective EHR with partially latent structure                              | Graph convolutional transformer                        | Learns hidden structure instead of relying on fixed meta-paths                 | Serious comparator because it questions whether meta-paths should be hand-designed at all           |
| [Jia23b] GraphCare            | Retrospective prediction with external KG and LLM-supported graph construction | Personalized knowledge graph plus BAT GNN              | Open-world knowledge augmentation                                              | Strong future-work direction and strong low-data comparator                                         |
| [Hu20] HGT                    | General heterogeneous graph method, not EHR-specific                           | Transformer style heterogeneous message passing        | Type-specific and edge-specific attention without manual meta-paths            | Strong methodological comparator if CareAI wants a more modern heterogeneous backbone               |
| [Xu23] Hypergraph Transformer | Retrospective EHR where visits contain joint code sets                         | Hypergraph plus self-attention                         | Higher-order relation modeling                                                 | Important if pairwise graph edges miss clinical code combinations                                   |
| [Che24b] TRANS                | Retrospective temporal heterogeneous graph across visits                       | Temporal graph transformer                             | Joint temporal and structural modeling with positional and structural encoding | Probably the strongest direct architectural challenge to static HAN++                               |
| [Hua24] HEART                 | Retrospective EHR pretraining across heterogeneous entities                    | Relation-aware heterogeneous transformer               | Pretrain then fine-tune with explicit pairwise relation embeddings             | Strongest signal that foundation style heterogeneous models are catching up fast                    |

| Study       | Study design and data view                                                           | Core technique                            | Analytical framework                                   | What it means for CareAI                                                             |
|-------------|--------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------|
| [Wan24] HAR | Retrospective diagnosis prediction with external knowledge graph and stage awareness | Hierarchical attentive relational network | Personalized sub-KG with disease-stage-aware attention | Good bridge between HAN ideas and richer knowledge-aware temporal diagnosis modeling |

## The model families compared

### Sequence and attention baselines

The sequence line treats EHR as a time-ordered visit history. Its strengths are maturity, simplicity, and natural handling of longitudinal data. Doctor AI, RETAIN, Dipole, and HiTANet all sit in this line [Cho15, Cho16b, Ma17, Luo20]. HiTANet is especially important because it shows that explicit time handling still matters even after basic attention is added [Luo20].

This family is weaker than graph methods when relations among diagnoses, tests, procedures, and medications are central. It is still a mandatory baseline family because a strong HAN++ paper must show that graph structure adds value beyond temporal sequencing.

### Knowledge-guided sequence models

GRAM and related models use ontology structure to stabilize rare-code learning and improve alignment with medical knowledge [Cho16]. These models are less expressive than a full heterogeneous graph, but they are easier to explain and harder to dismiss clinically. If CareAI is working with sparse labels or rare diseases, these baselines matter.

### Classical HAN line for clinical graphs

HeteroMed, HTAD, and later HAN-style EHR papers give the main methodological support for HAN++ [Hos18, Hos19, Qu22, Qu23]. Their core design is simple and attractive.

- represent EHR as a heterogeneous graph
- define clinically meaningful meta-paths
- use node-level attention within each semantic path
- use semantic-level attention across paths

This line supports the idea that different relation types matter differently for diagnosis and that hierarchical attention can provide more interpretable reasoning traces than flat sequence models [Hos19, Wan19].

Its main weaknesses are now clear.

- many studies are static or weakly temporal
- graph schema and meta-path choices are often hand-crafted and under-tested
- validation is mostly retrospective benchmark comparison
- explanation claims often rely on attention weights alone

The last point matters because general interpretability work warns that attention weights should not be treated as faithful explanation without extra tests [Jai19].

## Structure-learning and hub-aware heterogeneous graph models

This family is the most direct methodological challenge to plain HAN++.

GCT argues that hidden clinical structure can be learned instead of manually fixed through meta-path design [Cho19]. HSGNN argues that EHR graphs contain hub nodes that can distort naive heterogeneous attention and therefore need normalization and graph fusion [Liu20].

For CareAI, these papers imply that HAN++ should not merely inherit the standard HAN design. It should answer two questions:

- why are these meta-paths the right ones
- how is the model protected from common but weakly informative hubs

If those questions are unanswered, reviewers can argue that the graph design, not the architecture, produced the gains.

## Foundation and pretraining models

BEHRT, Med-BERT, and HEART change the comparison standard because they gain power from scale and transfer learning [Li19, Ras20, Hua24]. These models often do well when the local task dataset is not huge, and they can reduce the edge that handcrafted graph inductive bias once had.

This does not make HAN++ obsolete. It changes the burden of proof. HAN++ must show either:

- better task performance under the same data regime
- better explanation and controllability
- better integration of heterogeneous clinical relations
- better efficiency in smaller settings

## Personalized knowledge graph models

GraphCare and HAR extend the graph line by importing outside biomedical structure and patient-specific context [Jia23b, Wan24]. This family is important because it attacks two weak points in plain HAN++.

- fixed graph schemas may be too local
- patient context may need external knowledge to support rare or sparse cases

This is one of the most promising future directions for CareAI, especially if the system aims to recommend confirmatory tests or reason over disease mechanisms.

## Temporal graph and hypergraph models

TRANS and hypergraph transformers represent the strongest modern challenge to HAN++ for EHR prediction [Che24b, Xu23]. TRANS combines visit-to-visit time with graph structure, while hypergraph models capture higher-order code interactions that pairwise graphs can miss.

These models matter because many clinical states are not just paths between two node types. They are bundles of codes, labs, procedures, and visit transitions. A plain HAN++ model can miss that structure.

## Where each approach wins and loses

| Family                                  | Best feature                                                      | Main weakness                                            | CareAI verdict                                  |
|-----------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------|
| RNN and sequence attention              | Strong temporal ordering                                          | Weak relational structure                                | Must include as baseline, not final destination |
| Ontology-guided models                  | Good rare-code support and easier interpretation                  | Coarse structure                                         | Valuable comparator and ablation target         |
| Plain HAN style heterogeneous graphs    | Good semantic relation modeling and intuitive path attention      | Often static, hand-crafted, and weakly validated         | Keep as backbone, but do not stop here          |
| Hub-aware and structure-learning graphs | Better robustness to noisy graph construction                     | Less intuitive than plain HAN                            | Strong extension path for HAN++                 |
| Foundation and pretrained transformers  | Strong transfer and large-scale representation learning           | Less transparent and may ignore explicit graph semantics | Mandatory stress-test baseline                  |
| Personalized knowledge graph models     | Best for sparse settings and richer clinical context              | More system complexity and harder curation               | Strong future extension                         |
| Temporal graph and hypergraph models    | Strongest modeling of longitudinal and higher-order EHR structure | Higher complexity and heavier engineering                | Most serious competitor to HAN++                |

## What the literature says about study design quality

Most architecture papers in this area share the same study design.

- retrospective observational EHR benchmarks
- internal train and test splits on public datasets such as MIMIC
- performance emphasis on AUROC, F1, precision at k, or recall at k
- limited external validation
- limited calibration and decision analysis

That means model architecture papers often differ more in representation design than in clinical evaluation quality. The validation literature makes clear that a strong CareAI paper should break from that pattern [Alb17, Ril21, Ram20, Cab21].

| Validation issue    | Common pattern in model papers | Stronger CareAI standard                                                                              |
|---------------------|--------------------------------|-------------------------------------------------------------------------------------------------------|
| Discrimination      | AUROC or F1 only               | Add calibration-in-the-large, calibration slope, PR metrics, and confidence intervals [Alb17, Col15c] |
| Clinical usefulness | Rarely addressed               | Add decision curve or net benefit analysis [Vic06, Vic16]                                             |

| Validation issue    | Common pattern in model papers              | Stronger CareAI standard                                                                           |
|---------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------|
| External validation | Often absent or small                       | Report limits honestly and size it for precision where possible [Ril21, Ram20]                     |
| Bias and reporting  | Underreported preprocessing and missingness | Use TRIPOD style reporting and PROBAST thinking [Col15c, Wol19]                                    |
| Uncertainty         | Sometimes simple score thresholding         | Show patient-level brittleness and evaluate deferral or clinician review if claimed [Dus19, Liu21] |
| Interpretability    | Attention weights shown as explanation      | Add faithfulness and stability tests because attention alone is insufficient [Jai19]               |

## Strongest path for the model team

### Recommended architecture path

The strongest model path for the current team is a temporal HAN++ system rather than a static HAN++ system.

A practical architecture would look like this.

#### 1. Graph backbone

Keep HAN++ for heterogeneous entity fusion and semantic attention over clinically chosen relation paths [Wan19, Hos19, Qu23].

#### 2. Temporal layer

Add an explicit visit-level temporal encoder so the model handles disease progression across time rather than only local heterogeneous structure [Luo20, Che24b].

#### 3. Graph robustness layer

Add hub-aware normalization or graph fusion ideas from HSGNN, or at least run strong ablations to show common nodes do not dominate the result [Liu20].

#### 4. Knowledge option

If rare diseases or sparse data are important, add a lightweight knowledge augmentation path inspired by GRAM or GraphCare [Cho16, Jia23b].

This gives CareAI a stronger answer to current literature than plain HAN++ alone.

### Recommended baseline set

The comparison suite should include at least one strong paper from each major family.

- Sequence baseline: [Cho15] or [Ma17]
- Interpretable sequence baseline: [Cho16b]
- Knowledge-guided baseline: [Cho16]
- Transformer baseline: [Li19]



- Pretrained foundation baseline: [Ras20] and, if feasible, [Hua24]
- Classical graph baseline: [Hos18]
- HAN style clinical baseline: [Hos19] or [Qu23]
- Structure-learning graph baseline: [Cho19]
- Hub-aware heterogeneous graph baseline: [Liu20]
- Temporal graph baseline: [Che24b]
- Higher-order graph baseline: [Xu23]
- Knowledge-augmented graph baseline: [Jia23b]

A paper that beats only older RNN baselines will look dated. A paper that stays competitive across this set will look serious.

## Recommended analytical framework

The analytical framework should combine architecture testing with clinical validation.

- Multi-seed comparison with mean and variance
- Ablations on node types, edge types, and meta-paths
- Calibration analysis alongside discrimination
- Net benefit analysis for clinically relevant thresholds
- Shift analysis if there is any external or later-time cohort
- Uncertainty linked to physician review only if the workflow is actually evaluated [Dus19, Liu21]

## What should not be claimed yet

The current literature does not justify a blanket statement that HAN++ is the best model for EHR records. It supports a narrower statement.

HAN++ is a strong and clinically meaningful modeling choice for heterogeneous EHR diagnosis tasks, especially when relation semantics and explanation matter, but it must now be tested against newer temporal, pretrained, and higher-order models under stronger validation standards [Bol24b, Che24b, Hua24, Xu23].

That is a defensible claim. The universal-best claim is not yet defensible.

## Bottom line for the team

If the team wants the strongest publication path, it should not throw HAN++ away. It should upgrade HAN++ into a better-tested and more modern system.

The best near-term path is:

- temporal HAN++ backbone
- hub-aware graph checks
- hard comparator suite
- calibration and net benefit evaluation
- careful explanation claims
- honest external validation framing

That path keeps your current model investment, improves the baseline story, improves the validation story, and gives the reviewers fewer obvious ways to say the paper is outdated or overstated.

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